Mini Project Submission: Digital Circuit Design Specification & Datapath Schematic

Course: Digital Logic & Computer Design (DLCD)

Project Title: Hardware Neural Network Accelerator

Selected Module: Conv PE (`conv_pe.v`) with Q7.8 MAC (`mac_q7_8.v`)

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1. Executive Summary & Module Purpose

In the custom digital neural network accelerator, the **Convolutional Processing Element (`conv_pe.v`)** executes the 2D spatial convolution for the frontend convolutional layer (Conv1).

A fully unrolled 2D convolution requires 9 parallel hardware multipliers per output channel, which demands substantial FPGA DSP and logic resources. To minimize area and routing complexity, this design implements a **9-tap time-multiplexed, weight-stationary sequential Multiply-Accumulate (MAC) processing element**.

Key Architectural Characteristics:

- **Arithmetic Precision:** Signed Q7.8 Fixed-Point (16-bit signed two's complement: 1 sign bit, 7 integer bits, 8 fractional bits; scale factor $S = 2^8 = 256$, dynamic range [-128.0, +127.996]).
- **Dataflow:** Weight-Stationary. The 9 filter weights for the 3 × 3 kernel are shifted into internal registers during configuration and held stationary while incoming streaming pixels are evaluated sequentially.
- **Cycle Budget & Throughput:** Exactly 9 clock cycles per 3 × 3 receptive field window; produces 1 valid output feature every 9 cycles with continuous streaming support.
- **Hardware Component Footprint (`conv_pe`):** 1 Signed 16 × 16 Multiplier (`mac_q7_8.v`), 1 Signed 16-bit Adder (`mac_q7_8.v`), 9 16-bit Weight Registers (`weights[0:8]` serial shift register; unreset), 1 16-bit Accumulator Register (`out_acc`), 1 16-bit 2-to-1 Feedback Multiplexer (`clr_acc` selector), 1 16-bit 9-to-1 Weight Multiplexer (addressed by `tap_idx`), 1 4-bit Modulo-9 Binary Up-Counter (`tap_idx` , 0 to 8, gated by `mac_en`), 1 4-bit Comparator (`tap_idx == 8`), 2-input AND gate with `mac_en` , and 1 1-bit Output Valid Flip-Flop (`valid_out`). *(Note: The full Conv1 array in `conv1_array.v` instantiates 4 of these `conv_pe` instances in parallel, one for each output channel).*

2. Port Interface & Signal Contract

Signal Name	Direction	Bit Width	Type	Active Level	Functional Description
`clk`	Input	1	Clock	Rising Edge	Primary system clock (synchronous domain).
`reset`	Input	1	Control	High (1)	Synchronous reset. Resets `tap_idx` to 0, `out_acc` to 0, and `valid_out` to 0. *(Weights are intentionally unreset to conserve routing resources).*
`load_weight`	Input	1	Control	High (1)	Active-high shift enable for serial weight loading. Shift order requires reverse sequence (W_8 first down to W_0 last) so that W_0 resides in `weights[0]`.
`weight_in`	Input	[15:0]	Data (Q7.8)	—	Serial 16-bit signed weight stream fed during offline configuration.
`mac_en`	Input	1	Control	High (1)	Clock-enable for MAC computation and tap counter advance. When low, `tap_idx` and `out_acc` hold their current states.
`clr_acc`	Input	1	Control	High (1)	Asserted during Tap 0 to clear accumulator feedback to `16'sd0` for starting a new convolution window.
`pixel_in`	Input	[15:0]	Data (Q7.8)	—	16-bit signed streaming pixel value from sliding-window line buffers.
`out_acc`	Output	[15:0]	Data (Q7.8)	—	16-bit signed accumulated convolution sum. Valid when `valid_out` is asserted.
`valid_out`	Output	1	Status	High (1)	1-cycle active-high pulse registered on the clock cycle following Tap 8, indicating that the completed 9-tap sum is valid on `out_acc`.

3. Register Transfer Level (RTL) Datapath Schematic

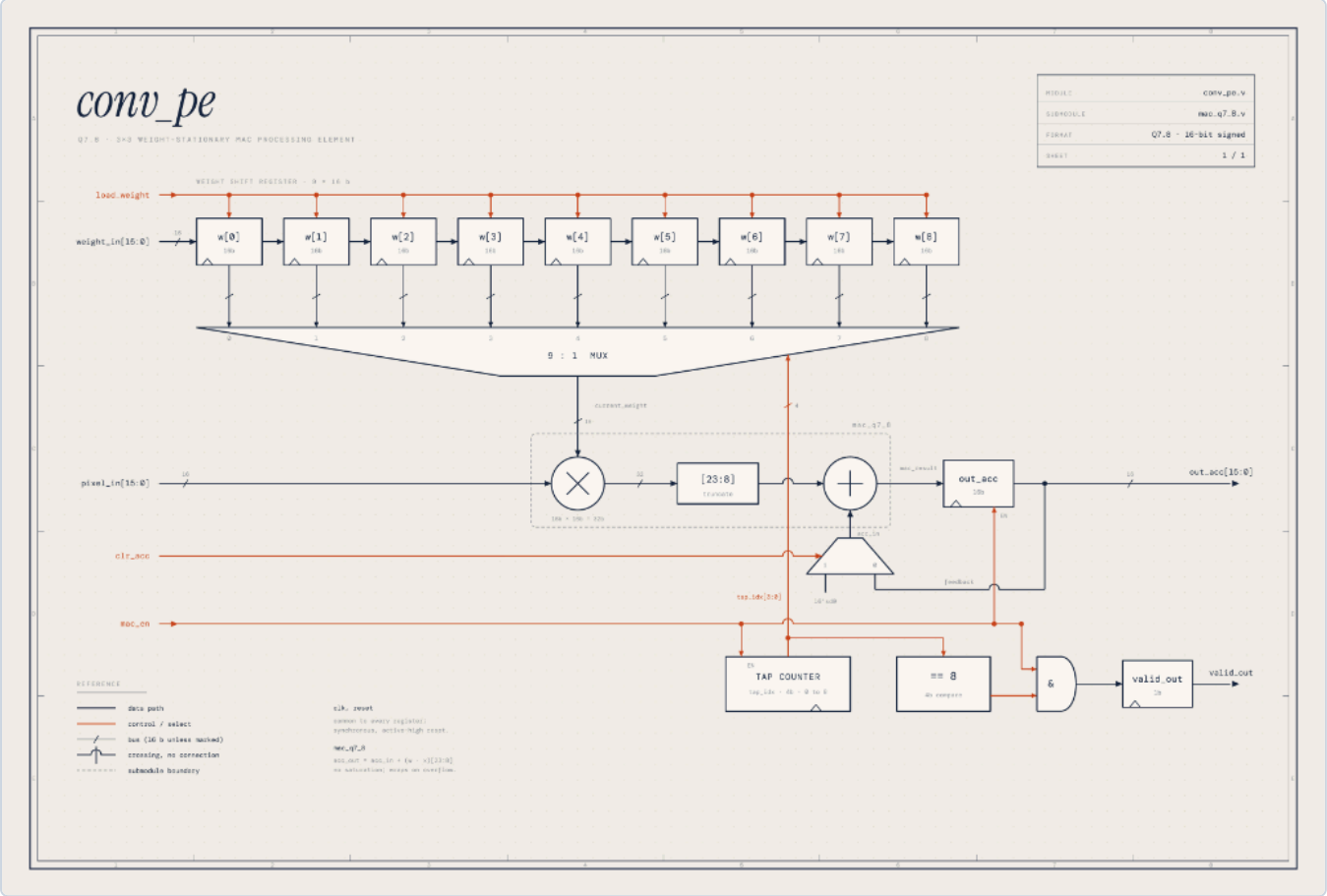


Figure 1: Complete Register-Transfer Level (RTL) Datapath Schematic of the Convolutional Processing Element (`conv_pe.v`) with Q7.8 MAC Core (`mac_q7_8.v`), Modulo-9 Tap Counter, and Valid Generation Logic.

Datapath Breakdown:

1. **Weight Storage & Selection:** The 9 kernel coefficients (W_0 to W_8) reside in a 9 × 16-bit shift register. A 9-to-1 multiplexer driven by `tap_idx[3:0]` routes `weights[tap_idx]` to the multiplier.
2. **Signed Multiplier (`mac_q7_8`):** Multiplies selected 16-bit signed weight and streaming 16-bit signed `pixel_in`, producing a 32-bit signed product in **Q15.16** format.
3. **Truncation Slice:** Combinational extraction of `product[23:8]`. Bits `[31:24]` are dropped based on verified headroom, and bits `[7:0]` are truncated (floor), aligning the result back to Q7.8 format at zero logic gate cost.
4. **Accumulator Feedback & MUX:** A 2-to-1 multiplexer selects `16'sd0` (when `clr_acc = 1` during Tap 0) or `out_acc` (when `clr_acc = 0`). The 16-bit signed adder sums the truncated product with this accumulator base.
5. **Output Registers:** When `mac_en = 1`, `out_acc` registers the adder sum on each clock edge. On `tap_idx == 8`, `valid_out` registers high for the subsequent clock cycle and `tap_idx` rolls over to 0.

4. Control Unit & Modulo-9 Tap Counter

The module avoids complex FSM control overhead by employing a synchronous 4-bit modulo-9 binary up-counter (`tap_idx`, counting 0 to 8) enabled by `mac_en`. When `mac_en = 0`, the counter freezes in place; it does not jump to an idle state. Weight loading is orthogonal via `load_weight`.

Modulo-9 Counter Signal Sequence Across a 9-Tap Window:

Cycle Phase	`mac_en`	`clr_acc`	`tap_idx`	Feedback MUX (`acc_in`)	Multiplier Operation (P_i)	`valid_out`
Tap 0	1	1	0	`16'sd0`	$P_0 = pixel_0 \times W_0$	0
Tap 1	1	0	1	`out_acc` (P_0)	$P_1 = pixel_1 \times W_1$	0
Tap 2	1	0	2	`out_acc` ($P_0 + P_1$)	$P_2 = pixel_2 \times W_2$	0
Tap 3..7	1	0	3..7	`out_acc` ($\sum_{k=0..i-1} P_k$)	$P_i = pixel_i \times W_i$	0
Tap 8	1	0	8	`out_acc` ($\sum_{k=0..7} P_k$)	$P_8 = pixel_8 \times W_8$	0
Handoff (Next Tap 0)	1	1	0	`16'sd0`	$P_0' = pixel_0' \times W_0$	**1** (Asserted for completed window)

5. Q7.8 Fixed-Point Mathematical Formulation & Headroom Analysis

Fixed-point arithmetic maps real values to signed integers using uniform scaling factor $S = 2^B$ (where $B = 8$): $X_{\text{fixed}} = \text{round}(X_{\text{real}} \times 2^8)$.

Multiplication & Bit Format:

Multiplying two 16-bit signed Q7.8 numbers produces a 32-bit signed product in **Q15.16** format:

> $\text{Product} = A \times B \;\Rightarrow\; 1 \text{ sign bit} + 15 \text{ integer bits} + 16 \text{ fractional bits} = 32 \text{ bits (Scale } 2^{16})$

To rescale to 16-bit Q7.8, the product is scaled by 2^{-8} via bit-slicing: **$\text{Product}_{\text{Q7.8}} = \lfloor (A \times B) / 2^8 \rfloor = \text{Product[23:8]}$**.

Bit-Level Slice of `product[31:0]`:

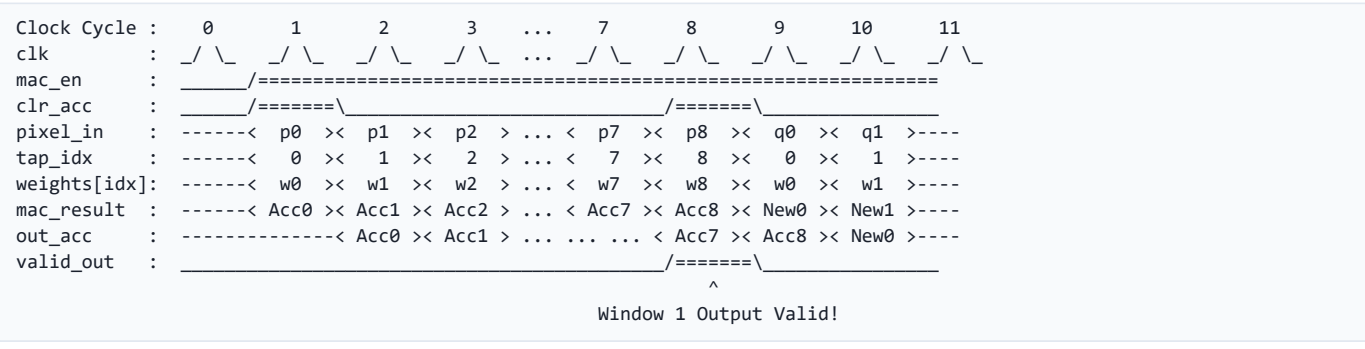
```
Product [31:0]:
  [31:24] (8 bits) : Extended integer bits (discarded; headroom bounded to prevent overflow)
  [23:16] (8 bits) : Q7.8 Integer part (1 sign bit + 7 integer bits)
  [15:8]  (8 bits) : Q7.8 Fractional part
  [7:0]   (8 bits) : Lower sub-fractional bits (truncated floor; dropped)
```

Headroom & Dynamic Range Quantification:

The RTL adder and accumulator implement standard two's complement 16-bit arithmetic without saturation logic; if a value exceeds [-128.0, +127.996], it wraps around. In the accelerator training pipeline, inputs are normalized with standard MNIST mean/std (μ = 0.1307, σ = 0.3081), producing pixel input values ranging from roughly -0.424 to +2.821. To rigorously verify that intermediate accumulations never overflow the 16-bit accumulator, the full test set was measured using the bit-exact integer emulator (`int_emulator_v2.py`):

- **Conv1 Accumulator Raw Range:** [-1253, +2080] integer counts.
- **Conv1 Accumulator Real Range:** [-4.895, +8.125] in Q7.8 format.
- **Peak Dynamic Range Utilization:** Peak magnitude is 8.125, consuming only **6.3%** of the available [-128.0, +127.996] dynamic range.
- **Headroom Margin:** 128.0 - 8.125 = 119.875 (> 14× safety margin), completely eliminating any risk of accumulator overflow.

6. Cycle-Accurate Timing Diagram & Simulation Waveform



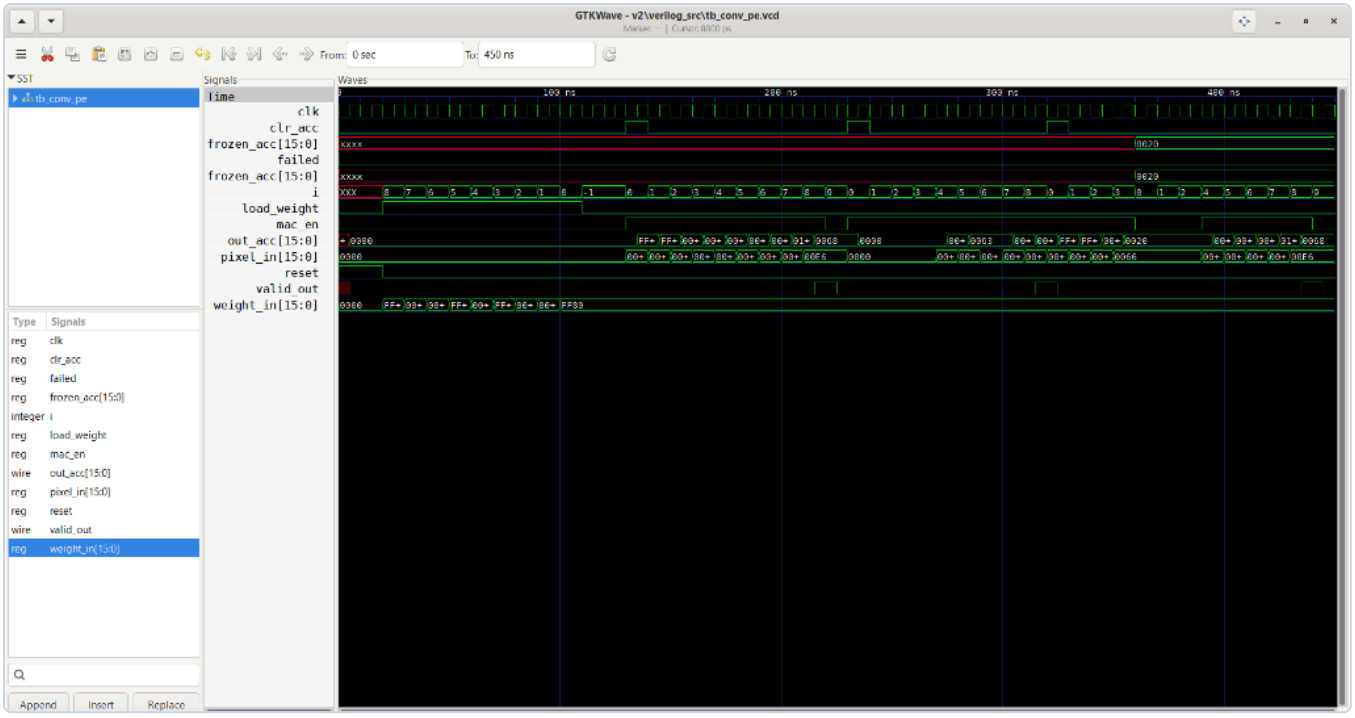


Figure 2: GTKWave Logic Simulation Trace of `tb_conv_pe.v` verifying Window 1 interior accumulation (`out_acc=104, valid_out=1`), Window 2 edge accumulation (`out_acc=19, valid_out=1`), and clock-enable gating.

7. Gate-Level Microarchitecture Schematic

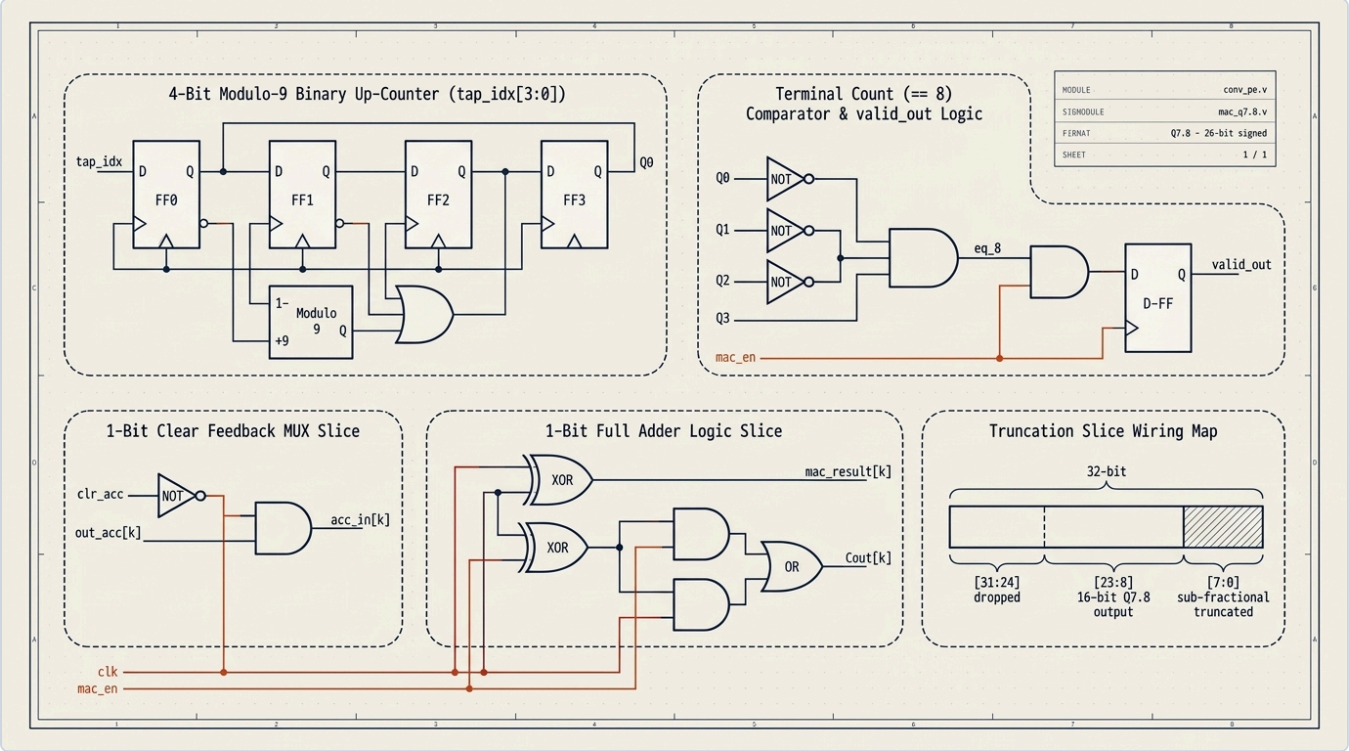


Figure 3: Gate-Level Logic Schematic illustrating (A) 4-bit Modulo-9 Synchronous Binary Up-Counter and Next-State Logic, (B) Terminal Count (`== 8`) Comparator and `valid_out` Register, (C) 1-Bit Clear Multiplexer Slice, (D) 1-Bit Full Adder Carry-Chain Slice, and (E) Truncation Wiring Slice.

Gate-Level Circuit Details:

1. **Modulo-9 Counter Logic:** Implemented using 4 D flip-flops with next-state combinational gates: $D_0 = \sim Q_0 \cdot \sim Q_3$; $D_1 = (Q_1 \oplus Q_0) \cdot \sim Q_3$; $D_2 = (Q_2 \oplus (Q_1 \cdot Q_0)) \cdot \sim Q_3$; $D_3 = (Q_3 \cdot \sim Q_0) + (Q_2 \cdot Q_1 \cdot Q_0)$.
2. **Terminal Count (`== 8`) Comparator:** Detects state 8 (1000b) via a 4-input AND gate: $eq_8 = Q_3 \cdot \sim Q_2 \cdot \sim Q_1 \cdot \sim Q_0$.
3. **Valid Output Gate:** A 2-input AND gate computes $D_{valid} = eq_8 \cdot mac_en$, clocking into the `valid_out` flip-flop on posedge `clk`.
4. **1-Bit Clear MUX Gate:** Implemented as $acc_in[k] = \sim clr_acc \cdot out_acc[k]$ (an inverter and AND gate), forcing `acc_in` to 0 on Tap 0.
5. **1-Bit Full Adder Slice:** Standard dual-XOR sum generation ($S_k = A_k \oplus B_k \oplus C_{in}$) and carry generation ($C_{out} = A_k \cdot B_k + C_{in} \cdot (A_k \oplus B_k)$).

8. Verilog Code Implementation

Core MAC Module (`mac_q7_8.v`):

```
module mac_q7_8 (
    input  wire signed [15:0] weight,
    input  wire signed [15:0] in_val,
    input  wire signed [15:0] acc_in,
    output reg  signed [15:0] acc_out
);
    wire signed [31:0] product;

    assign product = weight * in_val;
    always @(*) begin
        acc_out = acc_in + product[23:8];
    end
endmodule
```

Processing Element Module (`conv_pe.v`):

```
module conv_pe (
    input  wire clk,
    input  wire reset,
    input  wire load_weight,
    input  wire signed [15:0] weight_in,
    input  wire mac_en,
    input  wire clr_acc,
    input  wire signed [15:0] pixel_in,
    output reg  signed [15:0] out_acc,
    output reg  valid_out
);
    reg signed [15:0] weights [0:8];
    reg [3:0] tap_idx;
    wire signed [15:0] mac_result;
    wire signed [15:0] current_weight = weights[tap_idx];
    wire signed [15:0] acc_in = clr_acc ? 16'sd0 : out_acc;

    mac_q7_8 mac_inst (
        .in_val(pixel_in),
        .weight(current_weight),
        .acc_in(acc_in),
        .acc_out(mac_result)
    );

    always @(posedge clk) begin
        if (reset) begin
            tap_idx   <= 0;
            out_acc   <= 0;
            valid_out <= 0;
        end else begin
            if (load_weight) begin
                weights[8] <= weights[7];
                weights[7] <= weights[6];
                weights[6] <= weights[5];
                weights[5] <= weights[4];
                weights[4] <= weights[3];
                weights[3] <= weights[2];
                weights[2] <= weights[1];
                weights[1] <= weights[0];
                weights[0] <= weight_in;
            end
            if (mac_en) begin
                out_acc <= mac_result;
                if (tap_idx == 8) begin
                    tap_idx   <= 0;
                    valid_out <= 1;
                end else begin
                    tap_idx   <= tap_idx + 1;
                    valid_out <= 0;
                end
            end else begin
                valid_out <= 0;
            end
        end
    end
endmodule
```

9. Verification Results & FPGA Resource Analysis

1. Directed Testbench Verification (`tb_conv_pe.v`):

The module was verified using the self-checking testbench `v2/verilog_src/tb_conv_pe.v` compiled with Icarus Verilog:

- **Test Case 1 (Interior 3 × 3 Window):** Evaluated 9 non-zero pixel inputs (P_1 = [0.10, 0.20, ..., 0.90]) against signed kernel weights. Expected Q7.8 golden result: `104` (0.40625). RTL simulation output produced `104` and asserted `valid_out = 1` precisely on cycle 9 (`PASS`).
- **Test Case 2 (Edge 3 × 3 Window with Zero-Padding):** Evaluated boundary pixels with zero-padded corner and border positions (P_2 = [0, 0, 0, 0, 0.50, 0.60, 0, 0.80, 0.90]). Expected Q7.8 golden result: `19` (0.0742). RTL simulation output produced `19` and asserted `valid_out = 1` (`PASS`).
- **Test Case 3 (Mid-Stream `mac_en` Stall & Resume Test):** Deasserted `mac_en` for 3 clock cycles mid-stream at tap 4. Verified that `tap_idx` froze at 4, `out_acc` preserved its intermediate sum without corruption, and `valid_out` remained 0, before resuming to finish with `out_acc = 104` (`PASS`).
- **Summary:** Zero mismatches detected across directed test cases (`ALL TESTS PASSED.`).

2. Critical Path & FPGA Resource Mapping (Estimated Design Analysis, Unsynthesized):

- **FPGA Mapping Analysis:** Because the `[23:8]` truncation wiring slice sits between the multiplication and addition, the 16-bit adder cannot fold into a single DSP internal feedback accumulator. Consequently:
  - The 16 × 16 signed multiplier maps to a dedicated DSP slice (e.g. Xilinx DSP48E1).
  - The `[23:8]` bit-slicing and 16-bit signed adder map to the FPGA fabric logic using dedicated LUT carry chains (`CARRY4`).
- **Timing Estimate:** In typical FPGA fabrics (such as Xilinx Artix-7), the DSP multiplication combined with LUT carry-chain addition easily operates within a single clock cycle at estimated frequencies of 100 to 150 MHz without requiring multi-stage pipeline registers.

3. Hardware Resource Trade-Off Arithmetic:

Compared to a fully unrolled parallel 2D convolution engine that instantiates 9 multipliers per channel:

> Multiplier Reduction Ratio = (9 - 1) / 9 = 8 / 9 ≈ 88.89%

By time-multiplexing a single hardware multiplier across 9 clock cycles, the design reduces multiplier resource requirements by 88.89% per convolutional channel while maintaining deterministic 9-cycle throughput for real-time inference.